

Message Broker Function

The Message Broker Function enables publishing messages from DELMIA Apriso to an external message broker (Kafka or MQTT).

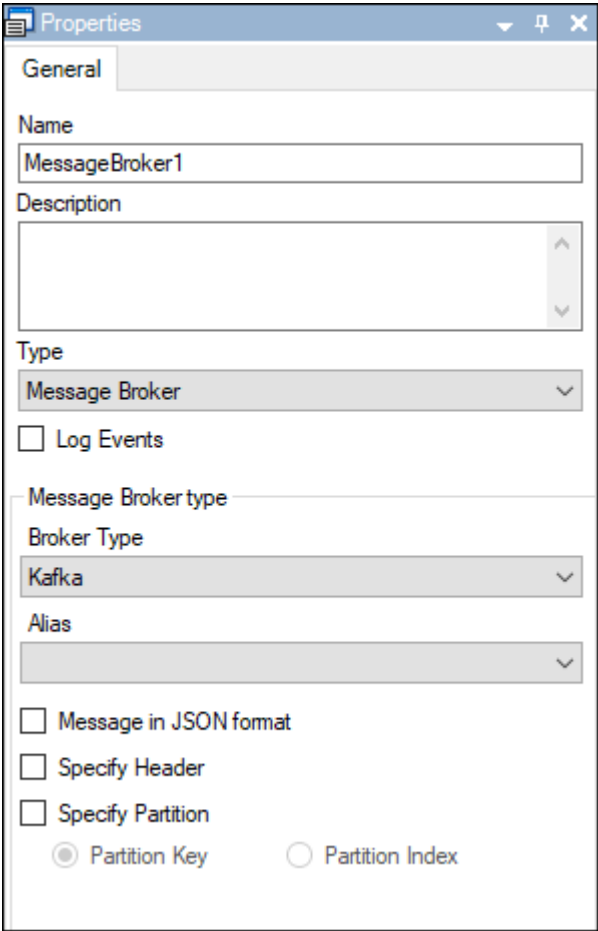
The function outputs a message to the broker in a list or JSON format.

Prerequisites

To use the Message Broker function, a connection to an external message broker must be configured in [Configuration Manager](#) .

To add the Message Broker Function

1. Drag the Message Broker Function from the [Toolbox](#) to the Function Navigation diagram, or double-click the Show Message icon in the Toolbox.



The screenshot shows the 'Properties' dialog box for the 'Message Broker' function. The 'General' tab is active. The 'Name' field is set to 'MessageBroker1'. The 'Description' field is empty. The 'Type' dropdown is set to 'Message Broker'. The 'Log Events' checkbox is unchecked. The 'Message Broker type' section shows 'Broker Type' set to 'Kafka' and 'Alias' set to an empty dropdown. The 'Message in JSON format' checkbox is unchecked. The 'Specify Header' checkbox is unchecked. The 'Specify Partition' checkbox is unchecked. The 'Partition Key' radio button is selected, and the 'Partition Index' radio button is unselected.

2. Choose the Broker Type (Kafka, MQTT, AMQPv9, or AMQPv10).
3. Select the message broker Alias.
4. Check **Message in JSON Format** to change the function's input format from a List to JSON.

Message Broker Function Inputs

Message in List Format

Input	Data Type	Description
Topic	Char	The topic of the message to be published. When using Kafka broker, the topic must be first defined on the broker side. Otherwise, the message will be ignored.
FieldList	List of Char	List of keys. The key names must be unique. The values of the duplicated keys are ignored.
ValueList	List of Char	List of key values.

Message in JSON Format

Input	Data Type	Description
Topic	Char	The topic of the message to be published. When using Kafka broker, the topic must be first defined on the broker side. Otherwise, the message will be ignored.
Message	Char	Message in JSON format.

Optional Inputs

Input	Data Type	Description
HeaderFieldList	List of Char	Available when you select the Specify Header checkbox. Must be unique. Used to create keys for header data, which is then passed to the published message. If specified, this input must be used together with HeaderValueList. Both inputs must have the same number of entries. To specify a header for an MQTT broker, you must first set the MQTT protocol to V5.0 for your alias in Configuration Manager . Otherwise, the provided input is skipped.
HeaderValueList	List of Char	Available when you select the Specify Header checkbox. Used to create values for header data, which is then passed to the published message.

		If specified, this input must be used together with HeaderFieldList. Both inputs must have the same number of entries. To specify a header for an MQTT broker, you must first set the MQTT protocol to V5.0 for your alias in Configuration Manager . Otherwise, the provided input is skipped.
PartitionKey	Char	Used only with a Kafka broker. Available when you select the Specify Partition checkbox. Used to specify the key of a partition to which the message is published.
PartitionIndex	Integer	Used only with a Kafka broker. Available when you select the Specify Partition checkbox. Used to specify the index of a partition to which the message is published.